

# $A \cup \phi = A$ $A \cap U = A$	Identity law	# $A \cap A = A$ $A \cup A = A$	Idempotent law
# $A \cup U = U$ $A \cap \phi = \phi$	Domination law	# $\overline{(\overline{A})} = A$	Complementation law
# $A \cup \overline{A} = U$ $A \cap \overline{A} = \phi$	Complement law	# $\overline{(A \cap B)} = \overline{A} \cup \overline{B}$ $\overline{(A \cup B)} = \overline{A} \cap \overline{B}$	De Morgan law
		# $\overline{A} \cap (A \cup B) = \overline{A}$ $A \cup (A \cap B) = A$	Absorption law

Ex Prove that $A - B = A \cap \overline{B}$

Pf let $x \in (A - B) \Leftrightarrow (x \in A) \wedge x \notin B$ ()
 $\Leftrightarrow (x \in A) \wedge (x \in \overline{B})$ ()
 $\Leftrightarrow x \in (A \cap \overline{B})$ (Defⁿ intersection)

$A - B = A \cap \overline{B}$

Ex let A, B, C be sets. $\overline{A \cup (B \cap C)} = (\overline{A} \cap \overline{B \cap C})$

Sol $\overline{A \cup (B \cap C)} = \overline{A} \cap \overline{(B \cap C)}$ (De Morgan law)
 ")

(h) $A \subseteq (A \cup B)$

Ex let $x \in A$
 $\Rightarrow (x \in A) \vee (x \in B)$ (Addition)

Sol ($\Rightarrow x \in (A \cup B)$

$A \subseteq (A \cup B)$

Ex 16 a) $(A \cap B) \subseteq A$ (b) $A \subseteq (A \cup B)$ (c) $(A - B) \subseteq A$

d) $A \cap (B - A) = \phi$ e) $A \cup (B - A) = A \cup B$

(a) $(A \cap B) \subseteq A$

let $x \in (A \cap B)$

$\Rightarrow (x \in A) \wedge (x \in B)$

$\Rightarrow x \in A$ (Simplification)

$(A \cap B) \subseteq A$

2.3 Functions

Set $\Rightarrow$ Cartesian Product $\Rightarrow$ Relation $\Rightarrow$ Function

Cartesian Product (Cross Product) | # Relation

Let A & B be sets

$$A \times B = \{(a, b) \mid a \in A, b \in B\}$$

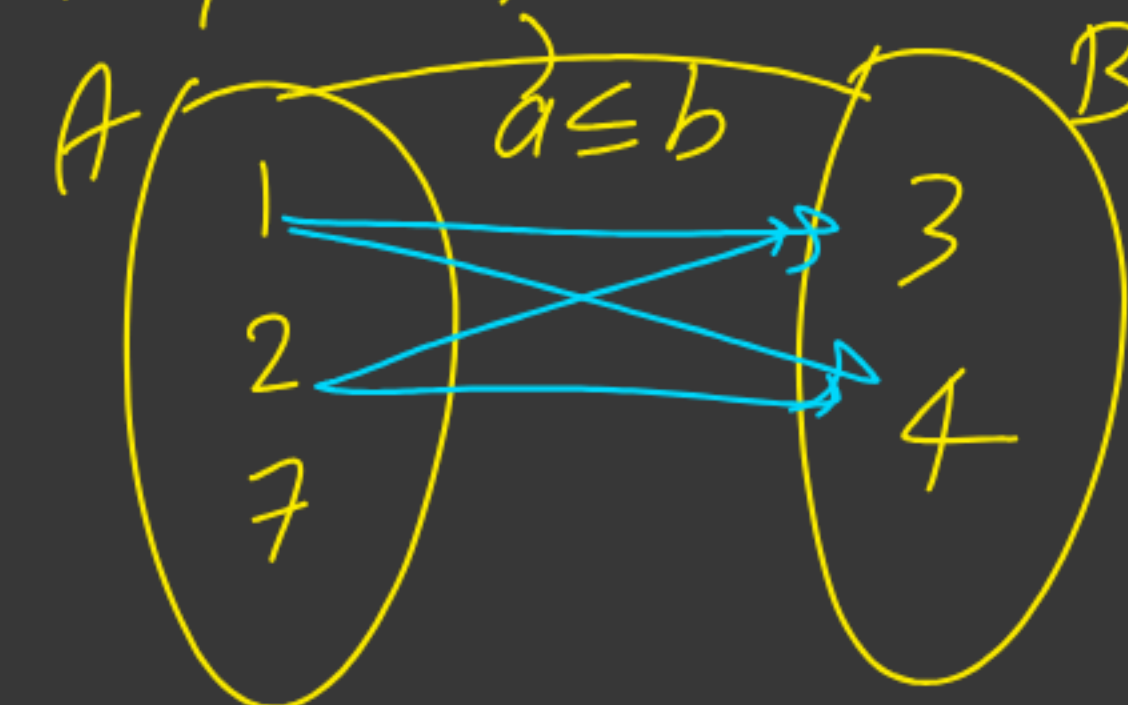
Ex: $A = \{1, 2\}$ $B = \{3, 4\}$

$$A \times B = \{(1, 3), (1, 4), (2, 3), (2, 4)\}$$

Ex: $A = \{1, 2, 7\}$ $B = \{3, 4\}$

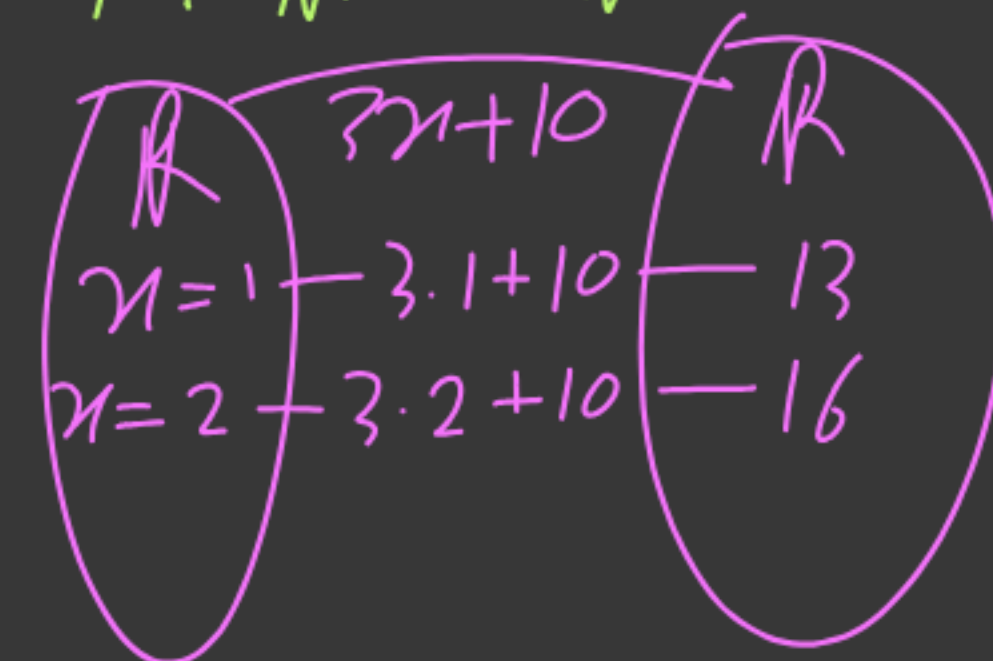
$$R = \{(a, b) \mid a \leq b, a \in A, b \in B\}$$

$$= \{(1, 3), (1, 4), (2, 3), (2, 4)\}$$



Ex: $f(x) = 3x + 10$

$$f: \mathbb{R} \rightarrow \mathbb{R}$$



① ✓

② ✓

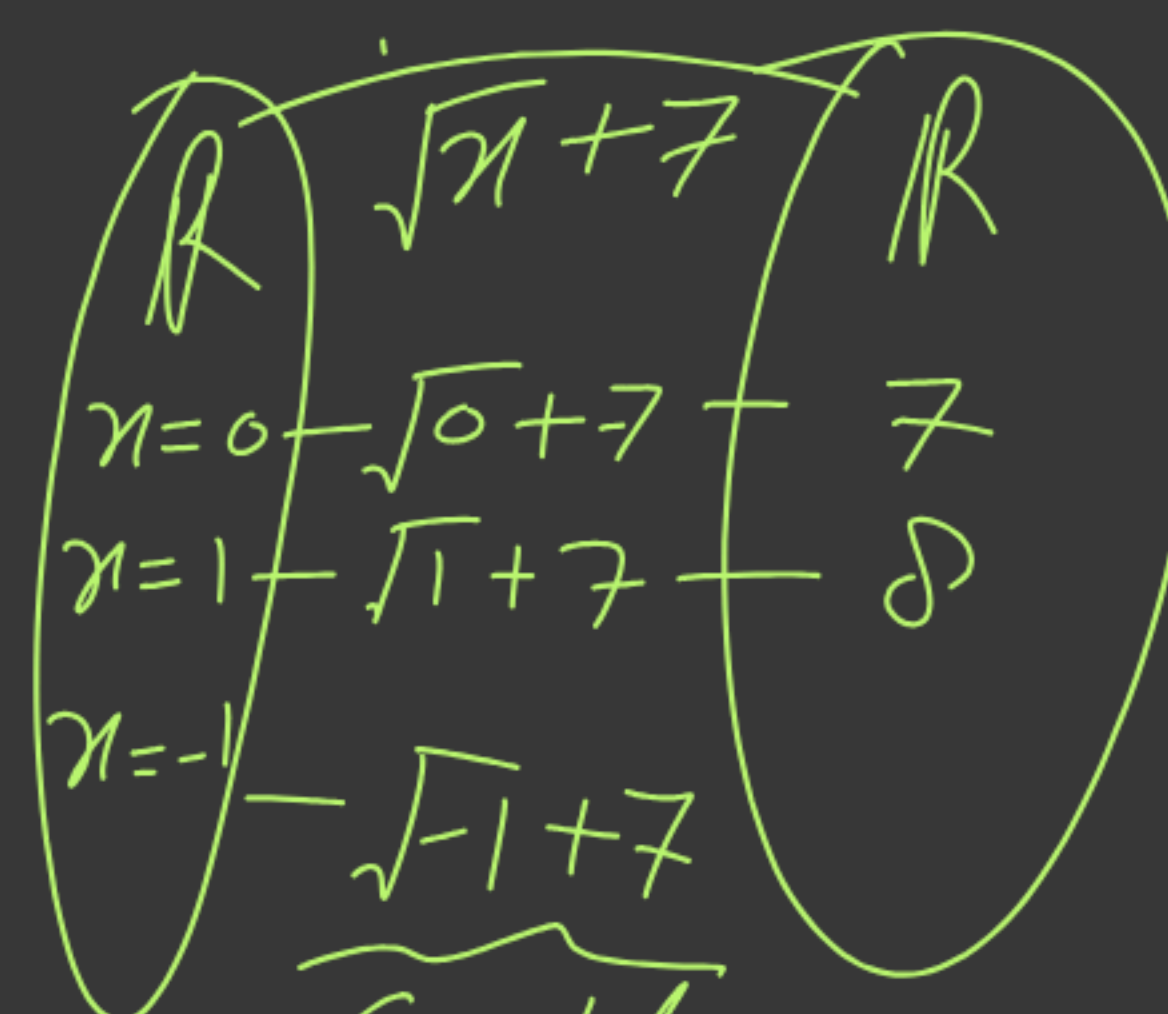
$$x_1 \rightarrow 3x_1 + 10 \rightarrow y_1$$

(X)

✓

Ex: $f(x) = \sqrt{x} + 7$

$$f: \mathbb{R} \rightarrow \mathbb{R}$$



Complex No.

① X

f is not a f

$$\text{Domain} = \mathbb{R} - (-\infty, 0) \quad \mathbb{R}^- \quad \mathbb{R}^+$$

$$\underline{\underline{f(c) \geq f(x), \forall x \in D.}}$$

Ch:-4. Applications of Differentiation.

4.1 Maximum and Minimum Values.

Max or Min	<u>$f(x)$</u>	Absolute	Max / Min $\rightarrow$ Global
		Local	Max / Min.
Domain $- D$			

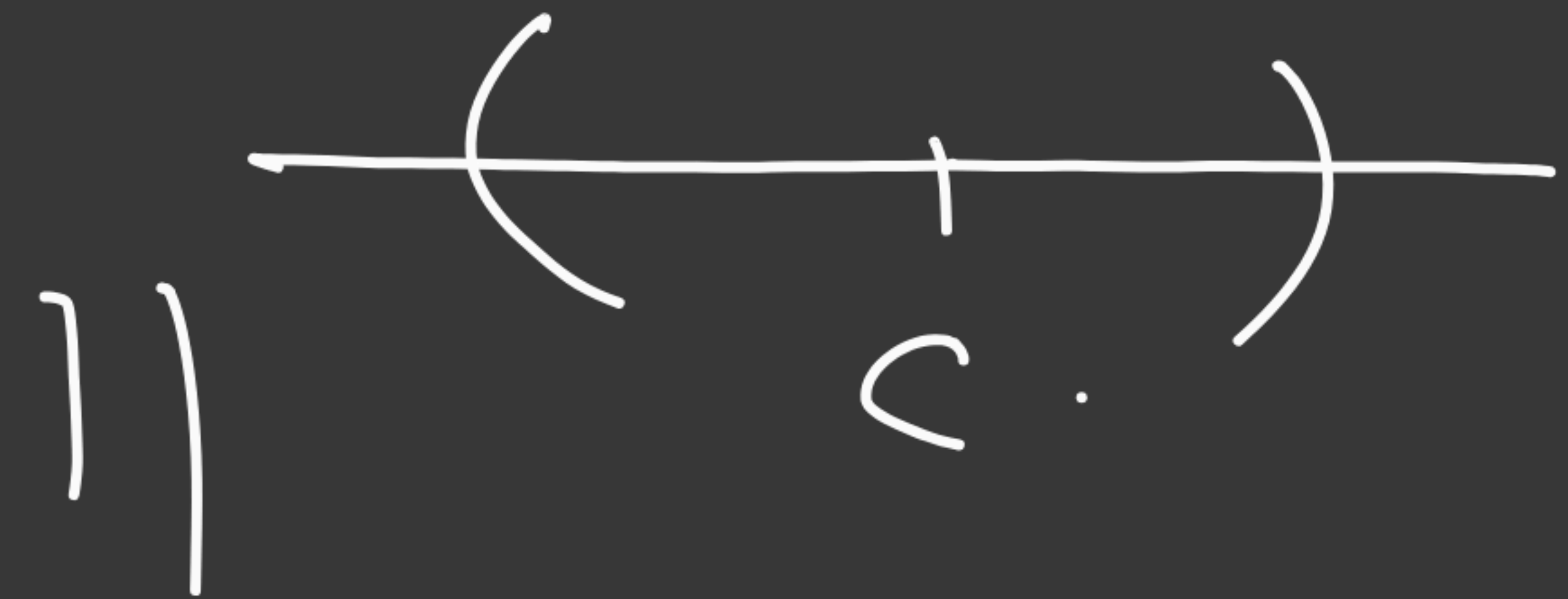
Let c be a no. belonging to the domain D of the funⁿ f and $f(c)$ is said to be the abs. max value of f if

min. $f(c) \geq f(x), \forall x \in D.$

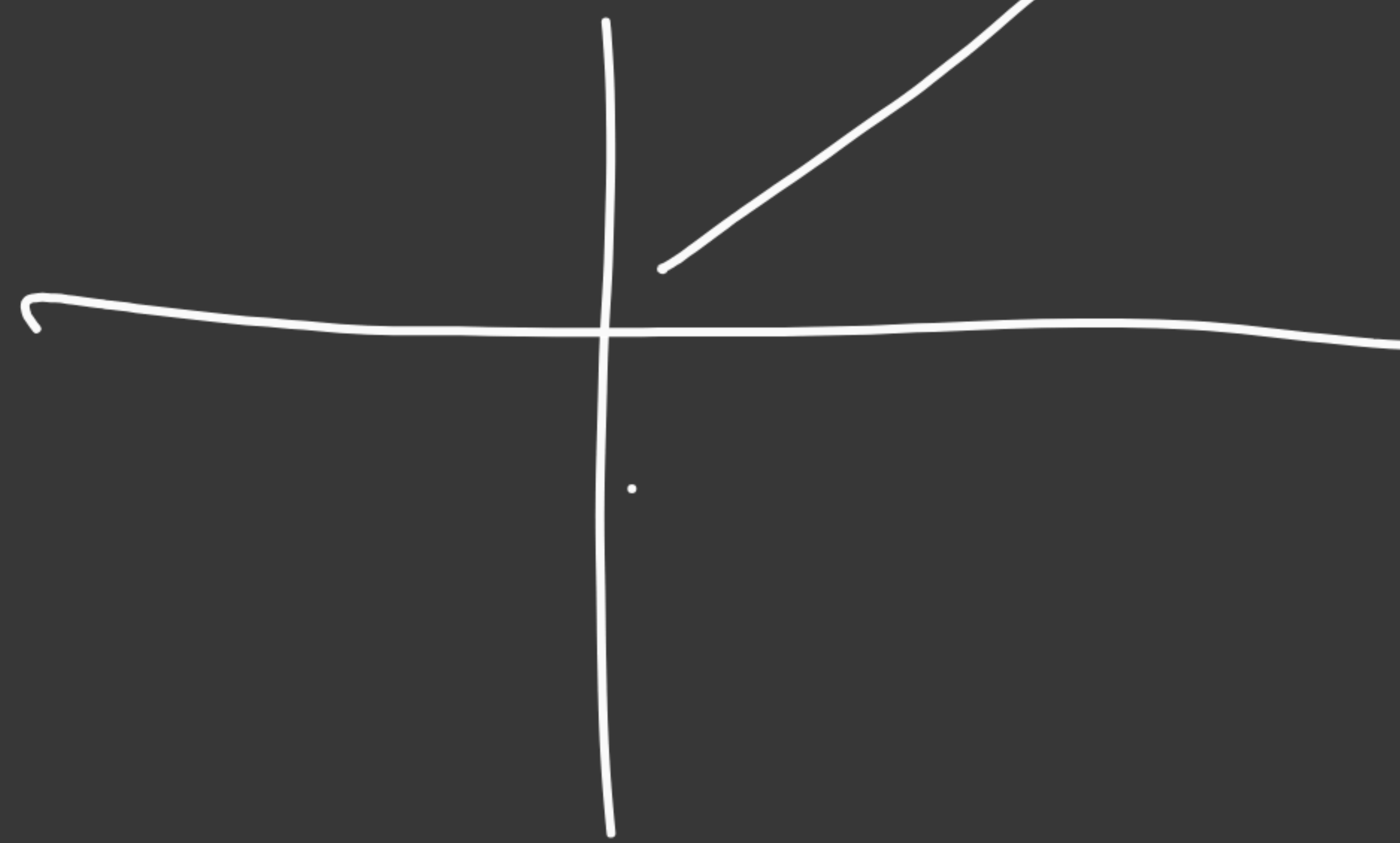
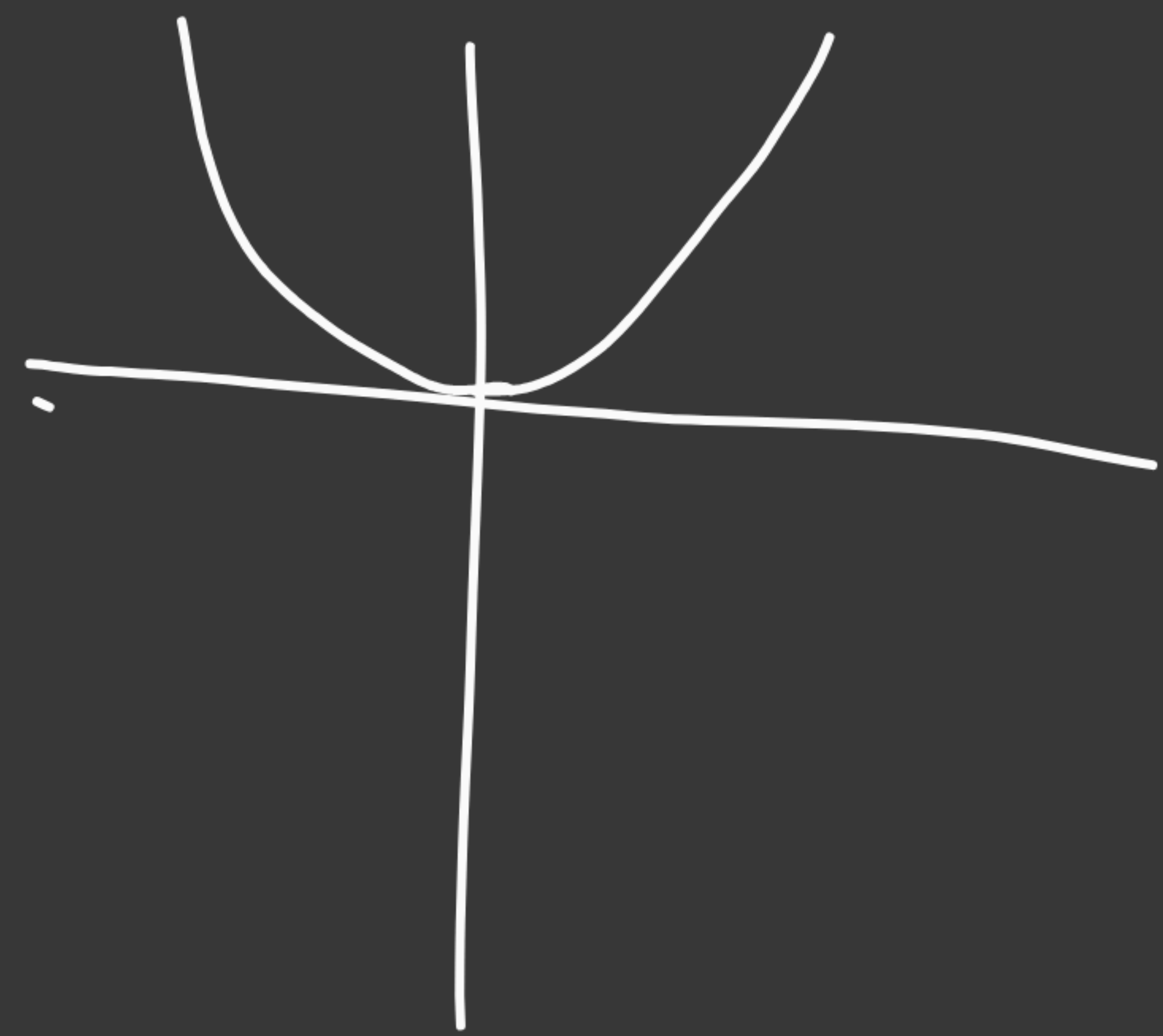
$\rightarrow f(c) \leq f(x), \forall x \in D.$

$f(c)$ is a local $\overset{\text{min}}{\text{max}}$ ^{value} of the funⁿ $f(x)$
if $\underline{f(c) \geq f(x)}$, for all x near c .

$$f(c) \leq f(x)$$



The Extreme Value Th:- If f is continuous on a closed interval $[a, b]$ then f attains an absolute maximum value $f(c)$ and an absolute minimum value $f(d)$ at some nos. c and d in $[a, b]$.

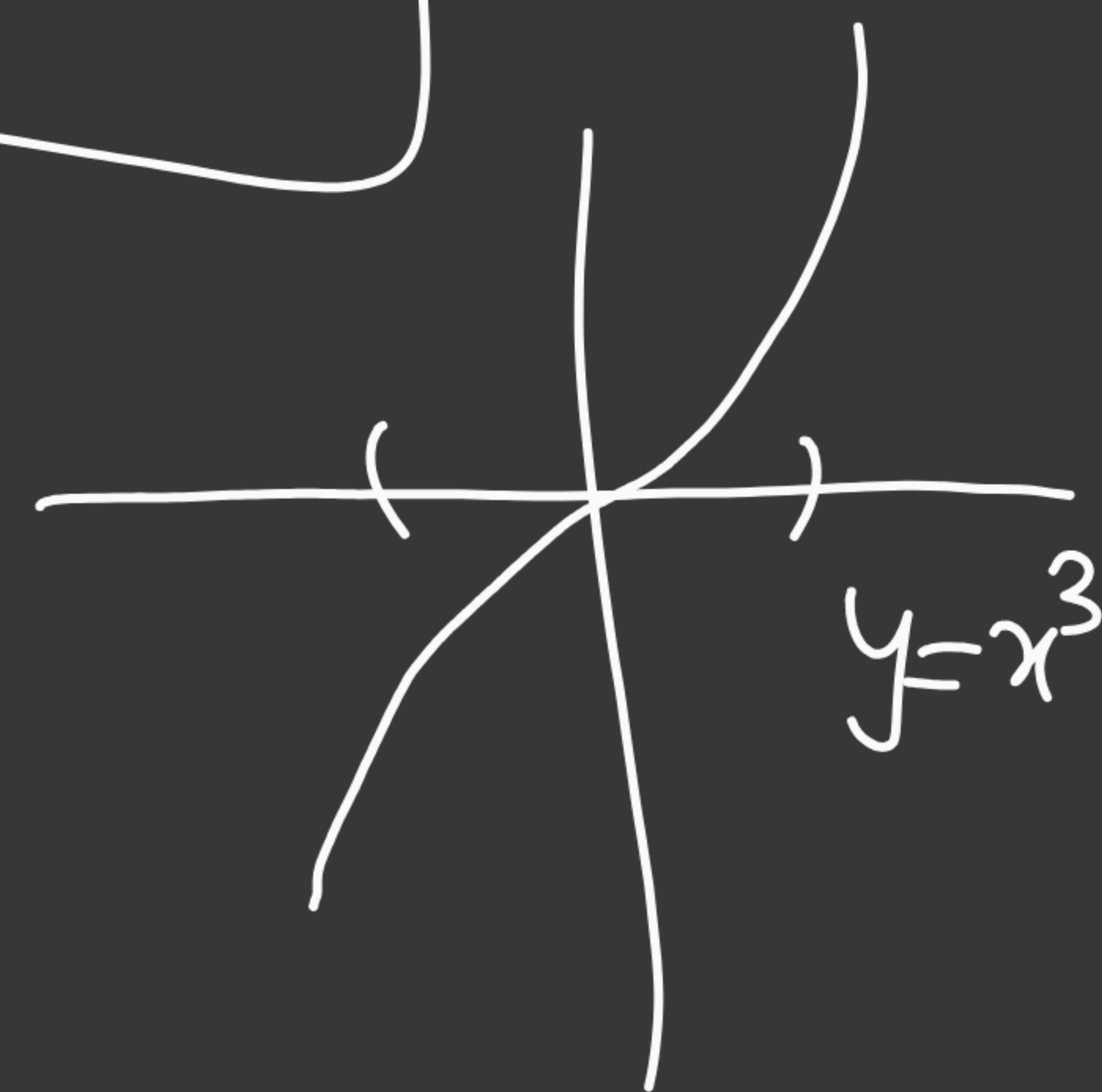


1. Fermat's Th:- If f has a local maximum/minimum at $\underline{c}$, and if $\underline{f'(c)}$ exists then $f'(c)=0$.

local max or min $\Rightarrow$ $f'(c)=0$

Ex: provided $f'(c)$ exists

$$f(x)=x^3 \Rightarrow f'(x) \Rightarrow 3x^2=0 \Rightarrow \boxed{x=0}$$



$$x^2 - x - 6 = 0 \Rightarrow x = -2, x = 3.$$

Defⁿ:- A critical number of a fuⁿ f is a no. c in the domain of the fuⁿ $f(x)$ such that either $f'(c) = 0$ or $f'(c)$ does not exist.

31) Find the critical nos. of the fuⁿ $f(x) = 2x^3 - 3x^2 - 36x$.

Solⁿ:- $f'(x) = 6x^2 - 6x - 36$
 Equating $f'(x) = 0$, $6x^2 - 6x - 36 = 0$.
 $\Rightarrow 6(x^2 - x - 6) = 0$.

$h(t)$ is not differentiable at $t=0 \Rightarrow t=0$ is a critical no.

37) $h(t) = t^{3/4} - 2t^{1/4}$.

$$h'(t) = \frac{3}{4} \frac{1}{t^{1/4}} - 2 \times \frac{1}{4} \frac{1}{t^{3/4}} = \frac{1}{4} \left[\frac{3}{t^{1/4}} - \frac{2}{t^{3/4}} \right]$$

$$= \frac{1}{4} \left(\frac{3t^{2/4} - 2}{t^{3/4}} \right) = \frac{1}{4} \left(\frac{3\sqrt{t} - 2}{t^{3/4}} \right)$$

Equating $h'(t) = 0 \Rightarrow \frac{1}{4} \left(\frac{3\sqrt{t} - 2}{t^{3/4}} \right) = 0 \Rightarrow 3\sqrt{t} - 2 = 0$
 $\Rightarrow \sqrt{t} = \frac{2}{3} \Rightarrow t = \frac{4}{9}$.

Defⁿ :— If f has a local maximum
or minimum at c , then c is a critical
no. of f .